LINJIE LYU

July 05, 1997
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https://linjielyu.github.io/

RESEARCH INTERESTS

- Diffusion Models Based 2D/3D Generation and Editing
- Global-illumination-aware Inverse Rendering, 3D Reconstruction, Relighting
- Radiance Fields, Efficient 3D Gaussian Splatting
- Uncertainty Quantification

EDUCATION

Max-Planck-Institut für Informatik, Saarbrücken, Germany — June 2021 — January 2026
Ph.D. Student in Computer Graphics and Vision.

- **Inverse Rendering & 3D Reconstruction:** Built frameworks to extract geometry, materials, and complex lighting from 2D images by integrating global illumination into differentiable pipelines.
- **Rendering Efficiency:** Replaced expensive Monte Carlo integration with highly efficient alternatives, including spherical harmonics for soft shadows and Neural Precomputed Radiance Transfer (PRT) fields.
- **Generative Model Integration:** Leveraged diffusion models (DDPMs) for ambiguity resolution in inverse problems and developed intrinsic-image latent space workflows for realistic scene editing.
- **Probabilistic Modeling:** Introduced uncertainty-aware rendering and differentiable manifold sampling to optimize subsequent data capture (next-best viewpoint planning).

Universität des Saarlandes, Saarbrücken, Germany — October 2019 — March 2021
Graduate School in Computer Science

Tsinghua University, Beijing, China — September 2015 — July 2019
Bachelor in Physics and Mathematics — * College Scholars; University Honors

Sichuan Mianyang High School, Mianyang, China — September 2012 — June 2015
* Silver Medal, Chinese Physics Olympiad

JOBS

Bridge (**LinkedIn**), Frankfurt, Germany — January 2026 — Present
Co-Founder & CTO
Co-founded and lead the technical development of an AI-powered buyer platform enabling Western companies to source directly from verified Chinese factories without the need for local teams. Architecting AI-driven solutions to streamline supplier selection, bridge complex technical communication gaps, and enforce rigorous quality control specifications.

INTERSHIP

Adobe, London, UK — June 2024 — September 2024
Research Intern During my research internship at Adobe, I developed a generative framework for pixel-precise image editing within an intrinsic-image latent space, culminating in our SIGGRAPH 2025 publication, IntrinsicEdit. This pipeline enables semantic, local manipulations—such as object insertion and removal, material adjustments, and scene relighting —while automatically resolving complex global-illumination effects like reflections and shadows.

Max-Planck-Institut für Informatik, Saarbrücken, Germany July 2018 — September 2018
CUDA-based visibility computation in Differentiable Rendering Tool (**Github**)
Engineered the core CUDA/C++ forward visibility and occlusion computation module for a custom differentiable rasterizer used in top-tier computer vision publications. Designed high-performance kernels to compute visibility independently of external graphics APIs (e.g., OpenGL) and integrated the module as a custom TensorFlow operator to support multi-view batch rendering and parametric shape fitting.

PUBLICATIONS

- **L. Lyu**, A. Tewari, J. Chen, T. Leimkühler, C. Theobalt. *Faster 3D Gaussian Splatting Convergence via Structure-Aware Densification*. Conference Proceedings, SIGGRAPH 2026
- X. Zhou, J. Chen, P. Rao, T. Teufel, **L. Lyu**, T. Minasian, O. Sotnychenko, X.-X. Long, M. Habermann, C. Theobalt. *OLATverse: A Large-scale Real-world Object Dataset with Precise Lighting Control*. CVPR 2026 (Oral).
- **L. Lyu**, V. Deschaintre, Y. Hold-Geoffroy, M. Hasan, J. Yoon, T. Leimkühler, C. Theobalt, I. Georgiev . *IntrinsicEdit: Precise Generative Image Manipulation in Intrinsic Space*. ACM Transactions on Graphics, SIGGRAPH 2025.
- **L. Lyu**, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, C. Theobalt. *Manifold Sampling for Differentiable Uncertainty in Radiance Fields*. Conference Proceedings, SIGGRAPH Asia 2024
- **L. Lyu**, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, C. Theobalt. *Diffusion Posterior Illumination for Ambiguity-aware Inverse Rendering*. ACM Transactions on Graphics, SIGGRAPH Asia 2023.
- **L. Lyu**, A. Tewari, T. Leimkühler, M. Habermann, C. Theobalt. *Neural Radiance Transfer Fields for Relightable Novel-view Synthesis with Global Illumination*. ECCV 2022 (Oral).
- **L. Lyu**, M. Habermann, L. Liu, M. B. R, A. Tewari,C. Theobalt. *Efficient and Differentiable Shadow Computation for Inverse Problems*. ICCV 2021.

SERVICES

Reviewer
SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, 3DV

Teaching
Computer Vision and Machine Learning for Computer Graphics